Mathematical Concepts

for (MATH261) Discrete Mathematics

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The principal definitions and mathematical results of (MATH261) Discrete Mathematics are developed here, with proofs. Notation and mathematical writing follow *(MATH000) Mathematical Language*; selected worked problems appear separately in the midterm document.

Theorem 1. Pigeonhole Principle
If N objects are distributed among $k \geq 1$ boxes, at least one box contains at least $\lceil N/k \rceil$ objects. In particular, distributing $k+1$ objects among k boxes puts at least two objects in one box.

Proof. If every box contained fewer than $\lceil N/k \rceil$ objects, each would contain at most $\lceil N/k \rceil - 1 < N/k$ objects. The total would then be less than N, a contradiction. The particular case follows by setting $N = k+1$. □

Definition 2. Binary Relation
A *binary relation* R on a set X is a subset of $X \times X$. We write xRy to mean $(x, y) \in R$. The relation R is

(i) *reflexive* if xRx for every $x \in X$;
(ii) *symmetric* if xRy implies yRx for all $x, y \in X$;
(iii) *antisymmetric* if xRy and yRx together imply $x = y$;
(iv) *transitive* if xRy and yRz together imply xRz.

Symmetry and antisymmetry are different conditions; neither is the negation of the other.

Definition 3. Equivalence Relation and Equivalence Class
An *equivalence relation* $\sim$ on X is reflexive, symmetric, and transitive. For $x \in X$, its *equivalence class* is

$$[x]_\sim \coloneqq \{y \in X : y \sim x\}.$$

Definition 4. Partition
A *partition* of a set X is a family $\mathcal{P}$ of nonempty, pairwise disjoint subsets whose union is X. For $X = \varnothing$, the empty family is its unique partition.

Theorem 5. Equivalence Relations and Partitions
The equivalence classes of an equivalence relation on X form a partition of X. Conversely, every partition $\mathcal{P}$ of X induces an equivalence relation given by $x \sim y$ if and only if x and y belong to the same block of $\mathcal{P}$. These constructions are inverse to each other.

Proof. Reflexivity gives $x \in [x]_\sim$, so the classes cover X and are nonempty. If $z \in [x]_\sim \cap [y]_\sim$, then $x \sim z$ and $z \sim y$ by symmetry, hence $x \sim y$ by transitivity. Every member of $[x]_\sim$ is therefore equivalent to y, and conversely; thus $[x]_\sim = [y]_\sim$. Distinct classes are disjoint.

Conversely, every $x \in X$ lies in exactly one block of a partition. Belonging to the same block is reflexive, symmetric, and transitive. Its equivalence class at x is precisely the block containing x, establishing both inverse assertions. □

Definition 6. Partial Order, Poset, and Total Order
A *partial order* $\preceq$ on X is a reflexive, antisymmetric, and transitive relation. The pair $(X, \preceq)$ is a *partially ordered set* (*poset*). Two elements x, y are *comparable* if $x \preceq y$ or $y \preceq x$; otherwise they are *incomparable*. A partial order is a *total order* if every pair is comparable. We write $x \prec y$ if $x \preceq y$ and $x \neq y$.

Definition 7. Cover Relation and Hasse Diagram
For x, y in a poset, y *covers* x, written $x \lessdot y$, if $x \prec y$ and there is no z with $x \prec z \prec y$. A *Hasse diagram* of a finite poset places its elements as vertices, with higher elements drawn above lower elements, and draws a segment between x and y precisely when $x \lessdot y$. Reflexive loops, arrowheads, and edges implied by transitivity are omitted. The order is recovered from upward paths, including paths of length zero.

Definition 8. Extremal Elements
Let $(X, \preceq)$ be a poset and $A \subseteq X$. An element $m \in A$ is *maximal* in A if no $a \in A$ satisfies $m \prec a$, and *minimal* if no $a \in A$ satisfies $a \prec m$. An element $g \in A$ is *greatest* in A if $a \preceq g$ for every $a \in A$, and *least* if $g \preceq a$ for every $a \in A$.

Proposition 9. Greatest and Maximal Elements
A greatest (respectively, least) element is unique, if it exists, and is maximal (respectively, minimal). A finite nonempty poset has a maximal and a minimal element, but need not have a greatest or a least element.

Proof. If g, h are both greatest, then $g \preceq h$ and $h \preceq g$, so antisymmetry gives $g = h$. If $g \prec a$ for some a, greatestness also gives $a \preceq g$, again contradicting antisymmetry. The dual assertions follow similarly. In a finite nonempty poset, start at any element and move to a strictly greater element whenever possible. Finiteness forces termination at a maximal element. Reversing the order gives a minimal element. □

Definition 10. Sunflower
A family of distinct sets $A_1, \ldots, A_r$ is a *sunflower* with r petals if a set K exists such that $A_i \cap A_j = K$ for every $i \neq j$. The set K is the *core*; the sets $A_i \setminus K$ are pairwise disjoint *petals*. In a k-uniform family every member has cardinality k.

Lemma 11. Sunflower Lemma
Let $k, r \in \mathbb{N}$ and let $\mathcal{F}$ be a family of distinct sets, each containing exactly k elements. If

$$|\mathcal{F}| > k!(r-1)^k,$$

then $\mathcal{F}$ contains a sunflower with r petals.

Proof. Induct on k. For $k = 1$, more than $r - 1$ distinct singleton sets have pairwise empty intersections, so any r form a sunflower. The case $r = 1$ is immediate for every k: the inequality guarantees $\mathcal{F} \neq \varnothing$.

Assume $k \geq 2$ and $r \geq 2$, and take a maximal pairwise disjoint subfamily $\{F_1, \ldots, F_t\} \subseteq \mathcal{F}$. If $t \geq r$, its first r members form a sunflower with empty core. Otherwise $t \leq r - 1$. Put $U = F_1 \cup \cdots \cup F_t$, so $|U| \leq k(r-1)$. Maximality implies that every $F \in \mathcal{F}$ meets U. Counting pairs (u, F) with $u \in U \cap F$ shows that some $u \in U$ belongs to at least

$$\frac{|\mathcal{F}|}{|U|} > \frac{k!(r-1)^k}{k(r-1)} = (k-1)!(r-1)^{k-1}$$

members. Removing u from each such member produces distinct $(k-1)$-element sets. By induction these contain a sunflower with r petals and core K; reinserting u gives the required sunflower with core $K \cup \{u\}$. □

Definition 12. Ordered Selections and Permutations
For $0 \le r \le n$, an *ordered selection without repetition* of r elements from an n-element set is an injective map $[r] \to [n]$, after identifying the ground set with $[n]$. The number of such selections is denoted by ${}_nP_r$. A *permutation* of a set X is a bijection $X \to X$; the set of permutations of $[n]$ is denoted by S_n.

Proposition 13. Number of Ordered Selections
For $0 \le r \le n$,

$$ {}_nP_r = n^{\underline{r}} = \frac{n!}{(n-r)!}. $$

In particular, $|S_n| = {}_nP_n = n!$, *with* $|S_0| = 1$.

Proof. An ordered selection has n choices for its first entry, $n-1$ for its second, and $n-r+1$ for its last. Multiplication yields the stated falling factorial. When $r = n$, every injection $[n] \to [n]$ is a permutation. □

Notation 14. Permutation Notation
For $\pi \in S_n$, the symbol $\pi(i)$ denotes the image of i. Its *two-line notation* is

$$ \pi = \begin{pmatrix} 1 & 2 & \cdots & n \\ \pi(1) & \pi(2) & \cdots & \pi(n) \end{pmatrix}. $$

The composition $\sigma \circ \pi$ sends i to $\sigma(\pi(i))$, so the rightmost map acts first. The identity $\mathrm{id}_{[n]}$ fixes every element, and π^{-1} denotes the inverse permutation. The cycle $(i_1\ i_2\ \cdots\ i_\ell)$ sends i_j to i_{j+1} for $1 \le j < \ell$, sends i_ℓ to i_1, and fixes all elements not listed. A fixed point is a cycle of length one.

Proposition 15. Disjoint-Cycle Decomposition
Every permutation of $[n]$ *is a product of disjoint cycles, unique up to the order of the cycles and cyclic rotation of the entries within each cycle.*

Proof. For any $i \in [n]$, the sequence $i, \pi(i), \pi^2(i), \ldots$ eventually repeats because $[n]$ is finite. Injectivity forces the first repeated element to be i, producing a cycle. Repeating this construction on unused elements partitions $[n]$ into disjoint cycles. The cycles are exactly the orbits of π, so their underlying sets are unique, and the entries in each cycle are determined up to cyclic rotation. □

Definition 16. Combinations
An *r-combination* of an n-element set is an r-element subset. Its number is denoted by ${}_nC_r$ or $\binom{n}{r}$. We set $\binom{n}{r} = 0$ when $r < 0$ or $r > n$, for $n \in \mathbb{N}_0$.

Proposition 17. Number of Combinations
For $0 \le r \le n$,

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}. $$

Proof. Each r-element subset has exactly $r!$ orderings. Thus the ${}_nP_r$ ordered selections are partitioned into groups of $r!$ according to their underlying subsets, and $\binom{n}{r} = {}_nP_r/r!$. □

Proposition 18. Pascal's Identity
For $n \ge 1$ *and* $0 \le r \le n$,

$$ \binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}. $$

Proof. Fix an element x of an n-element set. An r-element subset either contains x, leaving $r-1$ elements to be selected from the remaining $n-1$, or excludes x, leaving all r elements to be selected. The cases are disjoint and exhaustive.

Algebraically, for $1 \le r \le n-1$,

$$\begin{aligned}\binom{n-1}{r-1} + \binom{n-1}{r} &= \frac{(n-1)!}{(r-1)!(n-r)!} + \frac{(n-1)!}{r!(n-r-1)!} \\ &= \frac{r(n-1)! + (n-r)(n-1)!}{r!(n-r)!} = \frac{n!}{r!(n-r)!}.\end{aligned}$$

For $r = 0$ and $r = n$, the identity follows from the convention for out-of-range binomial coefficients. □

Theorem 19. Binomial Theorem
For $n \in \mathbb{N}_0$ and commuting variables x, y,

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r.$$

Proof. In the expansion of n factors $(x+y)$, a monomial $x^{n-r}y^r$ is obtained by choosing the r factors from which y is taken. There are $\binom{n}{r}$ such choices.

The theorem also follows by induction on n. It holds for $n = 0$. Assuming the formula at n and multiplying by $x+y$ gives

$$\begin{aligned}(x+y)^{n+1} &= \sum_{r=0}^{n} \binom{n}{r} x^{n+1-r} y^r + \sum_{r=1}^{n+1} \binom{n}{r-1} x^{n+1-r} y^r \\ &= x^{n+1} + \sum_{r=1}^{n} \left(\binom{n}{r} + \binom{n}{r-1} \right) x^{n+1-r} y^r + y^{n+1} \\ &= \sum_{r=0}^{n+1} \binom{n+1}{r} x^{n+1-r} y^r,\end{aligned}$$

where the last equality uses Pascal's identity. □

Definition 20. Multinomial Coefficient
For $n_1, \dots, n_k \in \mathbb{N}_0$ satisfying $n_1 + \cdots + n_k = n$, the *multinomial coefficient* is

$$\binom{n}{n_1, \dots, n_k} := \frac{n!}{n_1! \cdots n_k!}.$$

Proposition 21. Multinomial Counting Formula
The number of ways to distribute n distinct objects into k labeled groups of prescribed sizes $n_1, \dots, n_k$ is $\binom{n}{n_1,\dots,n_k}$.

Proof. Order all n objects, then place the first n_1 in group one, the next n_2 in group two, and so on. Each assignment occurs exactly $n_1! \cdots n_k!$ times among the $n!$ orders, corresponding to reorderings within the groups. □

Theorem 22. Multinomial Theorem
For $k \in \mathbb{N}$, $n \in \mathbb{N}_0$, and commuting variables $x_1, \dots, x_k$,

$$(x_1 + \cdots + x_k)^n = \sum_{\substack{n_1,\dots,n_k \ge 0 \\ n_1+\cdots+n_k = n}} \binom{n}{n_1, \dots, n_k} \prod_{i=1}^{k} x_i^{n_i}.$$

Proof. Choosing one term from each of the n factors determines a monomial $x_1^{n_1}\cdots x_k^{n_k}$. The coefficient is the number of assignments of the factor positions to k labeled groups of sizes $n_1,\dots,n_k$, namely the multinomial coefficient.

For an inductive proof on n, the formula is immediate at $n=0$. Assume it holds at n. Multiplying by $x_1+\cdots+x_k$, the coefficient of $x_1^{n_1}\cdots x_k^{n_k}$, where $n_1+\cdots+n_k=n+1$, becomes

$$\sum_{\substack{1\le j\le k\\ n_j\ge 1}} \binom{n}{n_1,\dots,n_j-1,\dots,n_k} = \sum_{\substack{1\le j\le k\\ n_j\ge 1}} \frac{n!n_j}{n_1!\cdots n_k!} = \frac{(n+1)!}{n_1!\cdots n_k!}.$$

Thus every coefficient agrees with the multinomial formula at $n+1$. □

Definition 23. Combinations with Repetition
For $n\ge 1$ and $r\in\mathbb{N}_0$, a combination with repetition is a multiset of cardinality r drawn from n types. Its number is denoted by ${}_nH_r$ or $\left(\!\binom{n}{r}\!\right)$.

Theorem 24. Stars and Bars
For $n\in\mathbb{N}$ and $r\in\mathbb{N}_0$, the number of solutions to

$$x_1+\cdots+x_n=r,\qquad (x_1,\dots,x_n)\in\mathbb{N}_0^n,$$

is $\binom{n+r-1}{n-1}=\binom{n+r-1}{r}$. It is also the number ${}_nH_r$ of r-combinations with repetition from n types.

Proof. Represent the solution by x_1 stars, a bar, x_2 stars, and so on, ending with x_n stars. This gives a string of r stars and $n-1$ bars. Conversely, the numbers of stars between consecutive bars recover the x_i uniquely. Choosing the positions of the $n-1$ bars among $n+r-1$ positions gives the formula. Selecting x_i objects of type i is the same data as the solution vector. □

Theorem 25. Inclusion–Exclusion Principle
For finite sets $A_1,\dots,A_n$,

$$\left|\bigcup_{i=1}^n A_i\right| = \sum_{\varnothing\ne I\subseteq[n]} (-1)^{|I|+1}\left|\bigcap_{i\in I} A_i\right|.$$

Equivalently, if $A_i\subseteq U$ for a finite universe U, then

$$\left|U\setminus\bigcup_{i=1}^n A_i\right| = \sum_{I\subseteq[n]} (-1)^{|I|}\left|\bigcap_{i\in I} A_i\right|,\qquad \bigcap_{i\in\varnothing} A_i = U.$$

Proof. Fix $u\in U$ that lies in exactly t of the sets. If $t=0$, its contribution to the first right-hand side is zero. If $t\ge 1$, its contribution is

$$\sum_{j=1}^{t}(-1)^{j+1}\binom{t}{j} = 1-(1-1)^t = 1.$$

Thus each element of the union is counted exactly once. Subtracting the first identity from $|U|$ proves the complement form.

Alternatively, induct on n. The case $n=1$ is immediate. For $n\ge 2$, the two-set identity gives

$$\begin{aligned}\left|\bigcup_{i=1}^n A_i\right| &= \left|\bigcup_{i=1}^{n-1} A_i\right| + |A_n| - \left|\bigcup_{i=1}^{n-1}(A_i\cap A_n)\right| \\ &= \sum_{\varnothing\ne I\subseteq[n-1]} (-1)^{|I|+1}\left|\bigcap_{i\in I} A_i\right| + |A_n| \\ &\quad - \sum_{\varnothing\ne I\subseteq[n-1]} (-1)^{|I|+1}\left|A_n\cap\bigcap_{i\in I} A_i\right|.\end{aligned}$$

The first sum counts index sets not containing n. The term $|A_n|$ corresponds to $\{n\}$, and each summand in the final sum corresponds to $I \cup \{n\}$, with sign $(-1)^{|I|+2}$. These are all nonempty subsets of $[n]$, with exactly the desired signs. □

Proposition 26. Number of Surjections
Let $n, k \in \mathbb{N}_0$. For $k \geq 1$, the number of surjections $[n] \to [k]$ is

$$\#\operatorname{Surj}([n],[k]) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n.$$

Here $0^0 = 1$ when an empty domain is counted. For $n = k = 0$ there is one surjection $\varnothing \to \varnothing$; for $n > 0$ and $k = 0$ there are none.

Proof. For $k \geq 1$, there are k^n maps $[n] \to [k]$. Let A_i be those maps whose image misses i. For any $J \subseteq [k]$, a map missing every element of J has $(k - |J|)^n$ choices. Inclusion–exclusion counts the maps in none of the A_i, which are exactly the surjections. □

Definition 27. Euler's Totient Function
For $n \in \mathbb{N}$, Euler's totient function is

$$\varphi(n) \coloneqq \#\{k \in [n] : \gcd(k, n) = 1\}.$$

In particular, $\varphi(1) = 1$.

Theorem 28. Euler's Totient Formula
If $n \in \mathbb{N}$ has distinct prime divisors $p_1, \ldots, p_t$, then

$$\varphi(n) = n \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right) = n \prod_{\substack{p \mid n \\ p \text{ prime}}} \left(1 - \frac{1}{p}\right).$$

The empty product for $n = 1$ equals 1.

Proof. For $n \geq 2$, an integer in $[n]$ fails to be coprime to n precisely when it is divisible by at least one of $p_1, \ldots, p_t$. For a subset $I \subseteq [t]$, exactly $n / \prod_{i \in I} p_i$ integers in $[n]$ are divisible by all the primes indexed by I. By inclusion–exclusion, the number divisible by none is

$$\sum_{I \subseteq [t]} (-1)^{|I|} \frac{n}{\prod_{i \in I} p_i} = n \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right).$$

For $n = 1$, the definition and empty-product convention agree. □

Definition 29. Derangements
A *derangement* of $[n]$ is a permutation $\pi \in S_n$ with no fixed point, so $\pi(i) \neq i$ for every $i \in [n]$. Denote the number of derangements by D_n (also $!n$). The empty permutation is a derangement by convention, so $D_0 = 1$; furthermore, $D_1 = 0$ and $D_2 = 1$.

Theorem 30. Derangement Formula
For every $n \in \mathbb{N}_0$,

$$D_n = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

Proof. For each $i \in [n]$, let $A_i = \{\pi \in S_n : \pi(i) = i\}$. If $I \subseteq [n]$ has cardinality j, fixing all elements of I leaves $(n - j)!$ permutations. Inclusion–exclusion therefore gives

$$D_n = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)! = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

The formula also gives $D_0 = 1$. □

Theorem 31. Derangement Recurrences
For $n \geq 1$ and $n \geq 2$, respectively,

$$D_n = nD_{n-1} + (-1)^n, \qquad D_n = (n-1)(D_{n-1} + D_{n-2}),$$

with $D_0 = 1$ and $D_1 = 0$.

Proof. Subtract the formulas for D_n and nD_{n-1} in Theorem 30 to obtain

$$D_n - nD_{n-1} = n!\frac{(-1)^n}{n!} = (-1)^n.$$

Applying this identity at n and $n-1$ yields

$$\begin{aligned} D_n &= nD_{n-1} + (-1)^n \\ &= (n-1)D_{n-1} + \big((n-1)D_{n-2} + (-1)^{n-1}\big) + (-1)^n \\ &= (n-1)(D_{n-1} + D_{n-2}). \end{aligned}$$

There is also a combinatorial proof of the second recurrence. Fix $j = \pi(n) \neq n$, with $n-1$ possible choices. If $\pi(j) = n$, the cycle $(n\ j)$ can be removed, leaving a derangement of the other $n-2$ elements. Otherwise the cycle containing n has length at least three. Deleting n from it yields a derangement of $[n-1]$. Conversely, for a given derangement of $[n-1]$, insert n immediately before j in the cycle containing j; this uniquely reconstructs the second case. Each choice of j therefore permits $D_{n-2} + D_{n-1}$ permutations. □

Definition 32. Catalan Numbers
The *Catalan numbers* are

$$C_n \coloneqq \frac{1}{n+1}\binom{2n}{n}, \qquad n \in \mathbb{N}_0.$$

Thus $C_0 = 1$, $C_1 = 1$, $C_2 = 2$, $C_3 = 5$, and $C_4 = 14$.

Theorem 33. Catalan Words
For $n \in \mathbb{N}_0$, the number of words $a_1 \cdots a_{2n}$ with $a_i \in \{-1, 1\}$, exactly n entries of each sign, and

$$\sum_{j=1}^{i} a_j \geq 0 \qquad (1 \leq i \leq 2n)$$

is C_n.

Proof. The assertion for $n = 0$ concerns the single empty word. Let $n \geq 1$. There are $\binom{2n}{n}$ words with n entries of each sign. Call a word *bad* if some prefix sum is negative. Locate the first prefix whose sum is -1 and reverse the signs of all entries in that prefix. Since that prefix contains one more -1 than $+1$, the resulting word has $n+1$ positive entries and $n-1$ negative entries.

This map is a bijection: a word with $n+1$ positive entries first reaches prefix sum $+1$ at a unique position, and reversing the signs up to that position reconstructs a bad word. Hence the number of bad words is $\binom{2n}{n+1}$, and the desired number is

$$\binom{2n}{n} - \binom{2n}{n+1} = \left(1 - \frac{n}{n+1}\right)\binom{2n}{n} = C_n.$$

□

Corollary 34. Catalan Lattice Paths
The number of paths from $(0,0)$ *to* (n,n) *consisting of* n *right steps and* n *upward steps that never pass above the diagonal* $y = x$ *is* C_n.

Proof. Replace each right step by $+1$ and each upward step by -1. At every prefix, the partial sum equals the current horizontal coordinate minus the vertical coordinate. Thus remaining on or below the diagonal is equivalent to the condition in Theorem 33. □

Remark. The same word can be read as a balanced parenthesis string, with $+1$ an opening parenthesis and -1 a closing parenthesis. Nonnegative prefix sums guarantee that no closing parenthesis occurs before its matching opening parenthesis. This is a bijective interpretation, not a separate counting problem.

Theorem 35. Catalan Recurrence
With $C_0 = 1$, *for every* $n \geq 1$,

$$C_n = \sum_{j=0}^{n-1} C_j C_{n-1-j}.$$

Proof. Consider a nonnegative balanced ± 1 word of length $2n$. Let its first return to partial sum zero occur after $2(j+1)$ entries, where $0 \leq j \leq n-1$. Its initial block is a $+1$, followed by a nonnegative balanced word of length $2j$, followed by a -1. The remaining suffix is independently a nonnegative balanced word of length $2(n-1-j)$. For this fixed j there are $C_j C_{n-1-j}$ choices by Theorem 33. Summing over the unique first-return index proves the recurrence. □

Definition 36. Stirling Numbers of the First Kind
Let $c(n,k)$ (also $\left[{n \atop k}\right]$) be the number of permutations in S_n whose disjoint-cycle decomposition has exactly k cycles. This is the *unsigned Stirling number of the first kind*. The *signed Stirling number of the first kind* is

$$s(n,k) \coloneqq (-1)^{n-k} c(n,k).$$

The initial and boundary values are $c(0,0) = s(0,0) = 1$, $c(n,0) = 0$ for $n \geq 1$, and $c(n,k) = 0$ for $k > n$. The signed values follow from $s(n,k) = (-1)^{n-k} c(n,k)$.

Proposition 37. First-Kind Stirling Recurrences
For $n \geq 1$ *and* $1 \leq k \leq n$,

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),$$
$$s(n,k) = s(n-1,k-1) - (n-1)s(n-1,k).$$

Proof. In a permutation of $[n]$ with k cycles, either n is a singleton cycle or it is not. In the first case, removing (n) leaves a permutation of $[n-1]$ with $k-1$ cycles. In the second, remove n from its cycle. The remaining permutation has k cycles, and n can be reinserted immediately after any one of the $n-1$ other elements, giving $n-1$ possibilities. This proves the first recurrence. Multiplying it by $(-1)^{n-k}$ and using $s(n,k) = (-1)^{n-k} c(n,k)$ gives the second. □

Proposition 38. First-Kind Stirling Special Values
For $n \geq 1$, $c(n,n) = 1$ *and* $c(n,1) = (n-1)!$; *for* $n \geq 2$, $c(n,n-1) = \binom{n}{2}$.

Proof. Only the identity permutation has n cycles. An n-cycle can be written beginning with 1, followed by any ordering of the remaining $n-1$ elements, yielding $(n-1)!$ permutations. To obtain $n-1$ cycles, exactly one pair of elements must form a transposition, while all other elements are fixed; there are $\binom{n}{2}$ choices. □

Proposition 39. Falling-Factorial Coefficients
For $n \in \mathbb{N}_0$,

$$x^{\underline{n}} = x(x-1)\cdots(x-n+1) = \sum_{k=0}^{n} s(n,k) x^k,$$

with $x^{\underline{0}} = 1$.

Proof. The result is immediate for $n = 0$. If it holds for $n-1$, then

$$x^{\underline{n}} = (x - n + 1) \sum_{k=0}^{n-1} s(n-1, k) x^k.$$

The coefficient of x^k on the right is $s(n-1, k-1) - (n-1)s(n-1, k) = s(n, k)$ by Proposition 37. □

Definition 40. Stirling Numbers of the Second Kind
The *Stirling number of the second kind* $S(n, k)$ (also $\left\{ {n \atop k} \right\}$) is the number of partitions of an n-element set into exactly k nonempty, unlabeled blocks. We set

$$S(0, 0) = 1, \quad S(n, 0) = 0 \ (n \geq 1), \quad S(n, k) = 0 \ (k > n).$$

For $n \geq 1$, $S(n, 1) = S(n, n) = 1$.

Proposition 41. Second-Kind Stirling Special Value
For $n \geq 2$, $S(n, n-1) = \binom{n}{2}$.

Proof. A partition of $[n]$ into $n-1$ nonempty blocks consists of one two-element block and $n-2$ singleton blocks. Choosing the pair uniquely determines the partition. □

Theorem 42. Second-Kind Stirling Recurrence
For $n \geq 1$ *and* $1 \leq k \leq n$,

$$S(n, k) = S(n-1, k-1) + kS(n-1, k).$$

Proof. Consider the block containing n. If it is the singleton $\{n\}$, the other $n-1$ elements form $k-1$ blocks. Otherwise, first partition $[n-1]$ into k blocks and then insert n into exactly one of these k blocks. The two cases are disjoint and exhaustive. □

Theorem 43. Surjections and Stirling Numbers
For $n \in \mathbb{N}_0$ *and* $k \in \mathbb{N}$,

$$\#\operatorname{Surj}([n], [k]) = k!S(n, k), \qquad S(n, k) = \frac{1}{k!} \sum_{j=0}^{k} (-1)^j \binom{k}{j} (k - j)^n.$$

For $(n, k) = (0, 0)$, *the corresponding cardinality identity is* $1 = 0!S(0, 0)$.

Proof. The nonempty fibers of a surjection $f : [n] \to [k]$ form a partition of $[n]$ into k blocks. Conversely, labeling the blocks of such a partition bijectively by $[k]$ determines a surjection; there are $k!$ such labelings. Combining this counting identity with Proposition 26 gives the explicit sum. □

Remark. The identities in Proposition 37 and Theorem 42 have similar forms but count different objects: cycles of permutations versus blocks of set partitions. Their initial conditions and signs must not be interchanged.

Definition 44. Bell Numbers
The *Bell number* B_n counts all partitions of an n-element set. Hence

$$B_n = \sum_{k=0}^{n} S(n, k), \qquad B_0 = 1.$$

The first values are $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, and $B_5 = 52$.

Theorem 45. Bell Recurrence
For $n \geq 1$,

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k.$$

Proof. Fix one distinguished element x of an n-element set. If exactly $n-1-k$ of the remaining elements lie in the block containing x, choose them in $\binom{n-1}{n-1-k} = \binom{n-1}{k}$ ways. The other k elements may be partitioned in B_k ways. Every partition determines a unique k, so summing proves the recurrence. □

Theorem 46. Bell Finite-Sum Formula
For $n \in \mathbb{N}_0$,

$$B_n = \sum_{k=0}^{n} \frac{k^n}{k!} \left(\sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \right),$$

where $0^0 = 1$ for the term occurring at $n = k = 0$.

Proof. For $n \geq 1$, use the explicit formula in Theorem 43 and rearrange the finite sums:

$$\begin{aligned} B_n &= \sum_{m=0}^{n} \frac{1}{m!} \sum_{j=0}^{m} (-1)^j \binom{m}{j} (m-j)^n \\ &= \sum_{m=0}^{n} \sum_{k=0}^{m} \frac{(-1)^{m-k} k^n}{(m-k)!k!} \\ &= \sum_{k=0}^{n} \frac{k^n}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}. \end{aligned}$$

For $n = 0$, both sides equal 1. □

Definition 47. Linear Homogeneous Recurrence Relation
A *linear homogeneous recurrence relation of degree k with constant coefficients* is

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}, \qquad n \geq k,$$

where $k \geq 1$, $c_1, \ldots, c_k \in \mathbb{R}$, and $c_k \neq 0$. A *solution* is a sequence satisfying the relation at every indicated index. We may also consider complex-valued solutions, with the same coefficients.

Proposition 48. Linearity of Homogeneous Solutions
If (a_n) and (b_n) solve the same linear homogeneous recurrence, then $(\alpha a_n + \beta b_n)$ also solves it for any scalars α, β in the underlying field.

Proof. For each $n \geq k$, multiply the recurrence for a_n by α, the recurrence for b_n by β, and add. The result is

$$\alpha a_n + \beta b_n = \sum_{j=1}^{k} c_j (\alpha a_{n-j} + \beta b_{n-j}).$$

□

Proposition 49. Existence and Uniqueness from Initial Values
For any prescribed $a_0, \ldots, a_{k-1}$, a recurrence of degree k as above determines exactly one sequence $(a_n)_{n \geq 0}$.

Proof. For existence, compute a_k from $a_0, \ldots, a_{k-1}$, and then calculate each subsequent term from the preceding k values. For uniqueness, two solutions with the same initial values agree at indices $0, \ldots, k-1$; if their preceding k terms agree, the recurrence forces their next terms to agree. Induction completes the argument. □

Definition 50. Characteristic Equation

The *characteristic polynomial* of the recurrence $a_n = \sum_{j=1}^{k} c_j a_{n-j}$ is

$$p(r) \coloneqq r^k - c_1 r^{k-1} - \cdots - c_{k-1} r - c_k.$$

The equation $p(r) = 0$ is its *characteristic equation.*

Proposition 51. Characteristic-Root Solutions

For $r \in \mathbb{C} \setminus \{0\}$, the sequence $(r^n)_{n\geq 0}$ solves the homogeneous recurrence if and only if $p(r) = 0$.

Proof. Substitution gives

$$r^n - \sum_{j=1}^{k} c_j r^{n-j} = r^{n-k} p(r).$$

Because $r \neq 0$, the recurrence holds for all $n \geq k$ exactly when $p(r) = 0$. □

Theorem 52. Second-Order Recurrences with Distinct Roots

Suppose $c_2 \neq 0$ and $r^2 - c_1 r - c_2$ has two distinct roots $r_1, r_2 \in \mathbb{C}$. Then the solutions of

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} \qquad (n \geq 2)$$

are precisely

$$a_n = \alpha r_1^n + \beta r_2^n, \qquad \alpha, \beta \in \mathbb{C}.$$

For real initial values, the resulting sequence is real; nonreal roots occur in conjugate pairs.

Proof. Each (r_i^n) is a solution by the characteristic equation, and linearity gives every displayed combination. Conversely, given a_0, a_1, the equations

$$\alpha + \beta = a_0, \qquad \alpha r_1 + \beta r_2 = a_1$$

have the unique solution

$$\alpha = \frac{a_1 - a_0 r_2}{r_1 - r_2}, \qquad \beta = \frac{a_0 r_1 - a_1}{r_1 - r_2}.$$

The resulting combination agrees with the prescribed initial values, hence with the original solution by Proposition 49. □

Theorem 53. Second-Order Recurrences with a Repeated Root

Suppose $c_2 \neq 0$ and $r^2 - c_1 r - c_2 = (r - r_0)^2$. Then all solutions of the associated second-order recurrence are

$$a_n = (\alpha + \beta n) r_0^n, \qquad \alpha, \beta \in \mathbb{C}.$$

Proof. Since $r_0 \neq 0$, coefficient comparison gives $c_1 = 2r_0$ and $c_2 = -r_0^2$. Besides r_0^n, the sequence nr_0^n is a solution, because

$$2r_0(n-1)r_0^{n-1} - r_0^2(n-2)r_0^{n-2} = nr_0^n.$$

By linearity, every displayed combination is a solution. To match arbitrary initial values a_0, a_1, choose $\alpha = a_0$ and $\beta = a_1/r_0 - a_0$. Uniqueness of the recurrence then establishes that every solution has this form. □

Theorem 54. General Solution of a Constant-Coefficient Homogeneous Recurrence

Let $p(r) = r^k - c_1 r^{k-1} - \cdots - c_k$, where $c_k \neq 0$. Suppose its distinct complex roots are $r_1, \ldots, r_t$ with multiplicities $m_1, \ldots, m_t$, so $m_1 + \cdots + m_t = k$. The solutions of the recurrence are exactly

$$a_n = \sum_{i=1}^{t} P_i(n) r_i^n, \qquad \deg P_i < m_i,$$

where each P_i is a polynomial over $\mathbb{C}$. The k polynomial coefficients are uniquely determined by k initial values. For real recurrence coefficients and real initial values, the expression can be chosen in conjugate pairs to yield a real-valued sequence.

Proof. Let E be the forward shift operator on sequences, $(Ea)_n = a_{n+1}$, and let $\Delta P(n) = P(n+1) - P(n)$. The recurrence is equivalent to $p(E)a = 0$ (with the equation indexed starting at $n = 0$). For $r \neq 0$,

$$(E - r)\big(P(n)r^n\big) = r^{n+1}\Delta P(n).$$

Consequently, $(E-r)^m(P(n)r^n) = r^{n+m}\Delta^m P(n) = 0$ whenever $\deg P < m$. Since the factors of $p(E)$ commute, every sequence in the displayed form is a solution.

These k basic solutions are linearly independent. Indeed, suppose

$$\sum_{i=1}^{t} P_i(n) r_i^n = 0 \qquad (n \geq 0),$$

where $\deg P_i < m_i$. Fix i and apply

$$\prod_{j \neq i} (E - r_j)^{m_j}.$$

All terms except the ith disappear. On sequences $P(n)r_i^n$, each factor $E - r_j$, for $j \neq i$, transforms P into

$$(r_i - r_j)P + r_i \Delta P.$$

On the vector space of polynomials of degree less than m_i, this map is triangular in the monomial basis with nonzero diagonal $r_i - r_j$, hence invertible. Therefore the surviving term can vanish only if $P_i = 0$. As i was arbitrary, all P_i vanish.

Thus the k basic solutions have independent initial-value vectors $(a_0, \ldots, a_{k-1}) \in \mathbb{C}^k$: otherwise their linear combination would have k zero initial terms and hence be identically zero by Proposition 49. These k vectors form a basis of $\mathbb{C}^k$, so every choice of initial values determines exactly one linear combination of the proposed form. □

Definition 55. Linear Nonhomogeneous Recurrence Relation
A *linear nonhomogeneous recurrence relation of degree k with constant coefficients* is

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n), \qquad n \geq k,$$

where $c_1, \ldots, c_k$ are constants, $c_k \neq 0$, and F is a prescribed function. The recurrence obtained by setting $F = 0$ is the *associated homogeneous recurrence*. A *particular solution* is any one sequence satisfying the nonhomogeneous recurrence.

Theorem 56. Homogeneous Plus Particular Solution
If $a_n^{(p)}$ is a particular solution to a linear nonhomogeneous recurrence, then every solution is of the form

$$a_n = a_n^{(p)} + a_n^{(h)},$$

where $a_n^{(h)}$ is an arbitrary solution of the associated homogeneous recurrence.

Proof. Subtract the recurrence satisfied by $a_n^{(p)}$ from the recurrence satisfied by any other solution a_n. Their difference $b_n = a_n - a_n^{(p)}$ satisfies $b_n = \sum_{j=1}^{k} c_j b_{n-j}$ because the forcing terms cancel. Conversely, adding a homogeneous solution to the particular solution preserves the forcing term $F(n)$, yielding another solution. □

Proposition 57. Method of Undetermined Coefficients
Let $p(r)$ be the characteristic polynomial of a linear recurrence with constant coefficients, and suppose its forcing term is $F(n) = P_d(n)\lambda^n$, where $\lambda \in \mathbb{C} \setminus \{0\}$ and P_d is a polynomial of degree at most d. If λ is a root of p of multiplicity m (and $m = 0$ if it is not a root), a particular solution may be sought in the form

$$a_n^{(p)} = n^m Q_d(n) \lambda^n,$$

where Q_d is a polynomial of degree at most d with undetermined coefficients. For a sum of such forcing terms, particular solutions may be added.

Proof. Let E denote the forward shift, $(Ea)_n = a_{n+1}$, and let $\Delta Q(n) = Q(n+1) - Q(n)$. With the recurrence indexed from $n = k$, its operator form is

$$(p(E)a)_n = F(n+k) \qquad (n \geq 0),$$

not $p(E)a = F(n)$. For $a_n = \lambda^n Q(n)$, direct calculation gives

$$(p(E)a)_n = \lambda^n p(\lambda(1+\Delta))Q(n).$$

If λ has multiplicity m, write $p(\lambda(1+z)) = z^m h(z)$ with $h(0) \neq 0$. The operator $h(\Delta)$ is invertible on polynomials of bounded degree: its matrix in the monomial basis is triangular with nonzero diagonal $h(0)$. Furthermore, Δ^m maps polynomials $n^m Q_d(n)$ with $\deg Q_d \leq d$ bijectively onto polynomials of degree at most d, since the image of n^{m+j} has leading term $(m+j)!n^j/j!$. Consequently the coefficients of Q_d can be uniquely chosen to match $\lambda^k P_d(n+k)$ on the right. Superposition follows from linearity. □

Forcing term $F(n)$	Particular-solution trial
$P_d(n)$	$n^m Q_d(n)$, where m is the multiplicity of 1 as a characteristic root
λ^n $(\lambda \neq 0)$	$n^m C\lambda^n$, where m is the multiplicity of λ
$P_d(n)\lambda^n$ $(\lambda \neq 0)$	$n^m Q_d(n)\lambda^n$
Sum of terms of these forms	Add the corresponding particular-solution trials

Remark. The method specifies a finite-dimensional trial family, not a guess without justification. After finding its coefficients, add the general homogeneous solution from Theorem 54 and use the initial conditions. Here P_d is prescribed, whereas Q_d has coefficients to be determined. A constant or polynomial forcing term corresponds to $\lambda = 1$.

Definition 58. Linear Recurrences with Variable Coefficients
A *linear recurrence with variable coefficients* has the form

$$a_n = \sum_{j=1}^{k} c_j(n) a_{n-j} + F(n),$$

where $n \geq k$, $k \geq 1$, and the coefficients $c_j(n)$ may depend on n. Unlike the constant-coefficient case, its solutions are not generally determined by a single characteristic polynomial.

Proposition 59. Reduction by Rescaling
Suppose a recurrence can be written as

$$f(n)a_n = \sum_{j=1}^{k} c_j f(n-j) a_{n-j} + F(n),$$

where $c_1, \ldots, c_k$ are constants. Setting $b_n = f(n)a_n$ gives

$$b_n = \sum_{j=1}^{k} c_j b_{n-j} + F(n).$$

If $f(n) \neq 0$, one recovers $a_n = b_n/f(n)$; when f vanishes at an index, that index must be handled separately.

Proof. Substitute $b_{n-j} = f(n-j)a_{n-j}$ for $0 \le j \le k$. The resulting relation has constant coefficients. □

Proposition 60. First-Order Variable-Coefficient Recurrences
Let

$$a_{n+1} = f(n)a_n + g(n), \qquad n \in \mathbb{N}_0,$$

where a_0 is given. With the convention that empty products equal 1, the unique solution is

$$\boxed{a_n = a_0 \prod_{j=0}^{n-1} f(j) + \sum_{m=0}^{n-1} g(m) \prod_{j=m+1}^{n-1} f(j).}$$

This formula remains valid even if some $f(j) = 0$. If $f(j) \neq 0$ for $0 \le j < n$, it can also be written as

$$a_n = \left(\prod_{j=0}^{n-1} f(j)\right)\left(a_0 + \sum_{m=0}^{n-1} \frac{g(m)}{\prod_{j=0}^{m} f(j)}\right).$$

Proof. For $n = 0$, the first formula gives a_0. Suppose it holds at n. Multiplying by $f(n)$ and adding $g(n)$ yields

$$\begin{aligned} a_{n+1} &= a_0 \prod_{j=0}^{n} f(j) + \sum_{m=0}^{n-1} g(m) \prod_{j=m+1}^{n} f(j) + g(n) \\ &= a_0 \prod_{j=0}^{n} f(j) + \sum_{m=0}^{n} g(m) \prod_{j=m+1}^{n} f(j), \end{aligned}$$

which is the formula at $n+1$. Induction proves the first expression, and forward recursion gives uniqueness. When all factors are nonzero, factoring out the cumulative product gives the second expression. Equivalently, one may divide the recurrence by that product and telescope. □

Remark. The product–sum formula is preferable to a quotient formula when some coefficient vanishes. In particular, a recurrence with $f(0) = 0$ must not be solved by dividing by $f(0)$; one may use the first formula or begin the quotient argument after the exceptional index.

Definition 61. Ordinary Generating Function
For a sequence $(a_n)_{n\ge0}$ over $\mathbb{R}$ or $\mathbb{C}$, its *ordinary generating function* (OGF) is the formal power series

$$A(x) \coloneqq \sum_{n=0}^{\infty} a_n x^n.$$

The notation $[x^n]A(x)$ denotes the coefficient a_n. Unless convergence is explicitly asserted, an identity of generating functions is an identity of formal power series, and does not require assigning a numerical value to x.

Proposition 62. Ordinary Generating Function Operations
Let $A(x) = \sum_{n\ge0} a_n x^n$ and $B(x) = \sum_{n\ge0} b_n x^n$. Then

$$\begin{aligned} A'(x) &= \sum_{n\ge0} (n+1)a_{n+1}x^n, \\ A(x) + B(x) &= \sum_{n\ge0} (a_n + b_n)x^n, \\ A(x)B(x) &= \sum_{n\ge0} \left(\sum_{j=0}^{n} a_j b_{n-j}\right) x^n. \end{aligned}$$

In particular, $[x^n](AB) = \sum_{j=0}^{n} a_j b_{n-j}$ is the Cauchy product *formula.*

Proof. Formal differentiation gives $\sum_{n\geq 1} na_n x^{n-1}$, and reindexing yields the first equality. Addition is coefficientwise. In the product, a monomial x^n arises precisely by multiplying $a_j x^j$ by $b_{n-j}x^{n-j}$ for $j = 0, \ldots, n$, proving the third equality. Each coefficient of a product involves only finitely many summands, so no convergence assumption is needed. □

Definition 63. Generalized Binomial Coefficient
For $\alpha \in \mathbb{R}$ and $n \in \mathbb{N}_0$, define

$$\binom{\alpha}{0} \coloneqq 1, \qquad \binom{\alpha}{n} \coloneqq \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} = \frac{\alpha^{\underline{n}}}{n!} \quad (n \geq 1).$$

When α is a nonnegative integer, this agrees with the usual binomial coefficient and vanishes for $n > \alpha$.

Theorem 64. Generalized Binomial Theorem
For every $\alpha \in \mathbb{R}$,

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \qquad (|x| < 1).$$

The coefficient identity also holds as the formal binomial expansion of $(1+x)^\alpha$. For nonnegative integer α, the series terminates and is the ordinary Binomial Theorem.

Proof. Write $b_n = \binom{\alpha}{n}$ and $F(x) = \sum_{n\geq 0} b_n x^n$. The coefficients satisfy

$$(n+1)b_{n+1} = (\alpha - n)b_n.$$

The ratio test shows that F converges absolutely for $|x| < 1$ unless it terminates, in which case convergence is automatic. On this interval, termwise differentiation is valid. Comparing coefficients using the displayed recurrence gives

$$(1+x)F'(x) = \alpha F(x), \qquad F(0) = 1.$$

Since $1 + x > 0$ for $-1 < x < 1$,

$$\frac{d}{dx}\big((1+x)^{-\alpha}F(x)\big) = 0.$$

The quantity in parentheses is therefore constant and equals its value 1 at $x = 0$. Thus $F(x) = (1+x)^\alpha$ throughout $|x| < 1$. The coefficient recurrence also verifies the formal power-series identity, independently of convergence. □

Remark. For $m \in \mathbb{N}$, negative exponents give $\binom{-m}{n} = (-1)^n\binom{m+n-1}{n}$. The corresponding explicit expansions of $(1+x)^{-m}$ and $(1-x)^{-m}$ may be obtained from Theorem 64; their derivation as worked examples belongs in the midterm document.

Definition 65. Exponential Generating Function
For a sequence $(a_n)_{n\geq 0}$ over $\mathbb{R}$ or $\mathbb{C}$, its *exponential generating function* (EGF) is

$$\widehat{A}(x) \coloneqq \sum_{n=0}^{\infty} a_n \frac{x^n}{n!}.$$

Its coefficients recover the original sequence by $a_n = n![x^n]\widehat{A}(x)$. As for OGFs, formal power-series manipulations do not require convergence.

Proposition 66. Exponential Generating Function Operations
Let $\widehat{A}(x) = \sum_{n\geq 0} a_n x^n/n!$ *and* $\widehat{B}(x) = \sum_{n\geq 0} b_n x^n/n!$. *Then*

$$\widehat{A}'(x) = \sum_{n\geq 0} a_{n+1}\frac{x^n}{n!},$$
$$\widehat{A}(x) + \widehat{B}(x) = \sum_{n\geq 0}(a_n + b_n)\frac{x^n}{n!},$$
$$\widehat{A}(x)\widehat{B}(x) = \sum_{n\geq 0}\left(\sum_{j=0}^{n}\binom{n}{j}a_j b_{n-j}\right)\frac{x^n}{n!}.$$

Proof. Differentiation of $a_n x^n/n!$ gives $a_n x^{n-1}/(n-1)!$ for $n \geq 1$, yielding the first identity after reindexing. The second follows coefficientwise. For the third, the coefficient of x^n in the product is

$$\sum_{j=0}^{n}\frac{a_j}{j!}\frac{b_{n-j}}{(n-j)!} = \frac{1}{n!}\sum_{j=0}^{n}\binom{n}{j}a_j b_{n-j},$$

which gives the asserted EGF coefficient formula. □

Remark. The product rules for ordinary and exponential generating functions differ: ordinary products involve the convolution $\sum_j a_j b_{n-j}$, whereas exponential products involve the binomially weighted convolution $\sum_j \binom{n}{j} a_j b_{n-j}$. Deriving the generating functions of Catalan, derangement, Bell, and Stirling sequences constitutes separate worked applications of these rules.